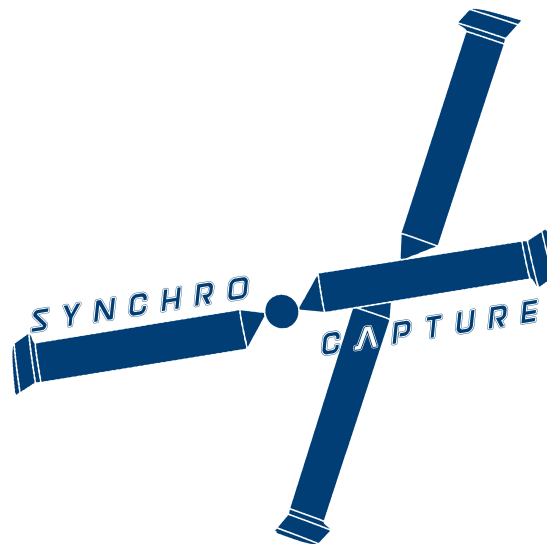


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

Fuselage Aerodynamics and Slung Load Modelling of a Novel Synchropter Concept for the Aerial-Recovery of Falling Rocket Stages



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Abstract

The fuselage sizing, optimisation as well as helicopter drag estimation with and without payload are presented in this report. The optimisation resulted in a fuselage with drag coefficient $C_D = 0.0314$ at zero incidence, lower than the analytical prediction of 0.0400 based on manned helicopters, as expected. This contributed to the total equivalent flat plate area in forward flight of $f = 0.418 \text{ m}^2$. The drag analysis with payload highlighted the large drag contribution of the first stage rocket due to its large dimensions with $L/D = 9.06$ with $C_D = 0.8109$. Lastly, the root NACA 23012 blade aerofoil was selected to minimise the likelihood of stall and extreme loads on the blade while maximising lift-to-drag performance.

WRR Team Members

Table 1: *Team Members and Roles.*

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra Atri Sharma Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston Hakan Serpen Tze Leung Wong	Flight Simulation & Testing	WRR06

†Sub-team leads are shown in **bold**

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Part I

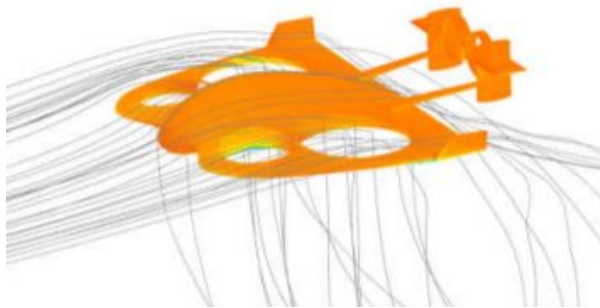
Pre-Conceptual

1 Air Vehicles Considered

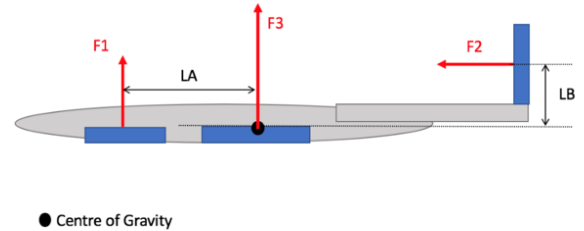
This project aimed at designing an air vehicle capable of capturing the first and second stage of a rocket when re-entering the atmosphere. This objective could be achieved with three different air vehicles: airplanes, helicopters or vertical take-off and landing (VTOL) aircraft. During the optimal vehicle research, the main constraint was the accuracy of re-entry provided by the launcher team. In fact, a large error would require more specific endurance and range characteristics than a small and accurate region of recovery where hover efficiency was potentially a more constraining design parameter. In this regard, four major design drivers were identified in order of importance: feasibility of design, likelihood of success, cost and time for capture. All four were evaluated for each vehicle by three separate teams.

1.1 VTOL

For the VTOL air vehicle three options were analysed by the business case team: conventional tilt-rotor, UAV tilt-rotor air crane and unconventional design with fan-in-wing concept. The main advantage of VTOL is the ability to cruise efficiently and to have a much larger range than helicopters although not as big as that of airplanes. While the first option was rather conventional as it resembled very closely to the V-22 Osprey or the AW609, the other two had new distinct features, namely being unmanned. Both were considered to be heavy-lifting and without a conventional bulky fuselage. The feasibility of a fan-in-wing unconventional VTOL was studied by myself and a sub-team: a similar pre-conceptual design is illustrated in a paper by Posva et al [2]. As seen in figure 1 the fuselage is a flying wing displaying 4 rotors for hover and pitch control and two fans at the tail for horizontal flight.



(a) Fan-in-wing VTOL concept [2]



(b) Pitch control moment arms and fan layout

Figure 1: VTOL design investigation

This configuration had the clear advantage of a large cruise efficiency and high L/D ratio as the fuselage generated the required lift while keeping the design very compact and manoeuvrable. This was thanks to the four rotors providing pitch control with moment arms L_A and L_B in figure 1b. Back-of-the-envelope calculations were then performed to understand the basics capabilities of this vehicle. In particular, the empty mass of 4500 kg for a payload of 1500 kg and a total required power of 6500 kW.

1.2 Air Vehicle Final Choice

Although the VTOL was not the final chosen configuration, the preliminary analysis was extremely informative to understand the crucial features the air vehicle must have: a large hover efficiency and lifting

capacity, given the first stage payload of 1,200 kg. **The final choice was the helicopter due to the expected accuracy of recovery provided by the launcher team of 50 km**, meaning that excellent cruise performance typical of aircraft was not as important as hover efficiency and manoeuvrability. Had the range been larger, a VTOL would have been a better choice given its ability to transition from hover to forward flight easily. However, due to the relatively short distance and high TRL (technology readiness level) the helicopter highlighted the best compromise for all design parameters.

2 Helicopter Baseline Configuration

The baseline configuration of the helicopter was chosen to comply with the mission profile of service ceiling of 20,000 ft and to facilitate operations and handling as much as possible. The four main features chosen are shown in figure 2: intermeshing blades, skids, H-tail and unmanned vehicle.

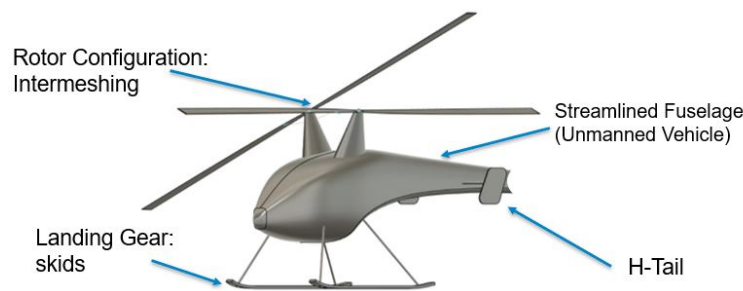


Figure 2: Helicopter baseline configuration

2.1 Intermeshing Rotor Blades

There were three possible choices regarding rotor blades: single main rotor, coaxial and intermeshing. Intermeshing rotors were chosen over coaxials due to larger hover efficiency and lower mechanical complexity: two separate rotors rather than two counter rotating rotors on a single axis. In addition, single main rotors were deemed less appropriate as they required a tail rotor and therefore a less compact design which could affect ground handling when taking-off from a boat.

2.2 Skids Landing Gear

Retractable landing gears were first discarded as the advantage of lower drag in forward flight was not worth the increased weight due to a heavier mechanism. In fact, due to the low operating range of 50 km from the boat's position, cruise performance was a less significant concern than hover and lifting capability. The choice was then between a wheel or skids configuration. However, due to the requirement of ground interference and the clearance of the intermeshing blades, tilted at an angle, skids were a more sensible choice over wheels. Skids also provide more stability of the vehicle when on the boat surface and have a lower drag coefficient than landing gears with wheels in cruise.

2.3 H-tail and Unmanned Vehicle

The H-shaped all moving horizontal tail with conventional rudders on the vertical stabilisers was selected and carefully sized considering the effect from the rotor's wake. This was an almost obvious choice due to the absence of a tail rotor. The unmanned vehicle design choice had the clear advantage of safety for pilots but also the significant reduction in fuselage size and constraints that cannot be neglected when

transporting passengers in the cockpit such as visibility. Therefore, this helped the optimal fuselage design and shape optimisation as a more streamlined shape could be considered. In this regard, the fuselage configuration and major aerodynamic surfaces sizing were the main responsibility in the latter part of the project concerning preliminary design.

Part II

Preliminary Design

3 Fuselage Sizing

3.1 Initial Sizing

The initial fuselage sizing, first iteration, consisted of finding regression data for a given MTOW and payload mass. Data was collected solely for similar intermeshing rotor configurations and also compared to helicopters with coaxial rotors (the R-squared values were 0.83 and 0.78 respectively for the length-MTOW relationship). The initial fuselage length from the nose to the tail excluding the rotor diameter was found to be in this range: $8.53 \text{ m} \leq L \leq 9.96 \text{ m}$. The width of the fuselage was estimated as 1.5 m from initial sizing information provided by the propulsion and systems team. However, given the exotic helicopter design, the length was just as an initial figure to be refined in the iterations as the reference data was all based on manned vehicles.

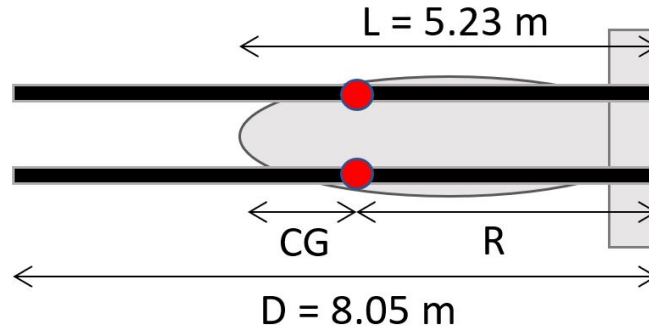


Figure 3: Second iteration of the fuselage sizing (view from above)

A crucial design choice strictly relevant to aerodynamic considerations has been made from the beginning: **the H-tail must always interact with the wake of the intermeshing rotors both in hover and forward flight.** This was done to fully optimise the tail in consistent flow conditions and avoid any trade-offs to satisfy when the flow features would have changed. Such decision was also significant to set a geometric constraint on the maximum length of the fuselage. Since the wake must interact with the tail in cruise and in hover, the maximum length was governed by two parameters. The former is the distance between the centre of gravity corresponding to the main rotor centre, illustrated with red circles, and the end of the tail boom, the latter was the CG position itself. In figure 3, a CG location of around $0.20L$ (from the nose) was initially assumed to establish a lower bound. As a consequence the maximum length could be expressed as a function of the diameter alone as shown in equation 1.

$$L_{\text{MAX}} = CG + \frac{D}{2} = 0.23L_{\text{MAX}} + \frac{D}{2} \quad (1)$$

The newly derived equation was purely dependent on the rotor diameter and indirectly by the CG, fixed by the systems internal layout. Although the second iteration was rather far from the first estimate with regression data, the result was judged reasonable as investigating similar helicopter configuration whose diameter is larger than the fuselage gave very similar geometry features. Notably, the fuselage length to rotor diameter ratio of 0.65.

The fuselage length was finalised in the third iteration when the exact CG location was published by the structures team [3], which appeared to be $0.40L$. Thanks to the assumptions made in the second iteration, a recalculation of the final length was possible yielding **6.60 m**, this number was above the lower range of the second iteration and below the upper range provided by the regression data from manned helicopters. In addition, the maximum width did not differ substantially from the first estimate as it was computed directly from the internal system layout. In conclusion, the final length and maximum width were confirmed to be:

$$L = 6.60 \text{ m} \quad w_{\max} = 1.70 \text{ m}$$

Such fixed measures were then used as the primary geometric constraints for the overall fuselage shape optimisation.

3.2 Fuselage Shape Optimisation

As a first step, literature research of previous optimisations was carried out to find the most effective and efficient procedure given the time constraints and the purpose of the design project. In specific, a helicopter fuselage testing wind tunnel NASA report [4] was studied to highlight the major aerodynamic parameters. Since the simulation was performed with CFD, papers concerning rear fuselage optimisation [5] and good practices for fuselage drag computational estimation were consulted [6] [7].

3.2.1 Optimisation Geometry Constraints and Parameters

After the basic internal system disposition of components was completed by the structures, propulsion and systems teams, the fuselage shape could be designed. Although the helicopter operated at a maximum speed of 225 km/h, which is low compared to airliners, drag played an important role in helicopters and can be up to an order of magnitude larger than aircraft [8]. For this reason, the shape optimisation was crucial especially in cruise to optimise the performance of the synchropter. In order to to this, two major sets of parameters were clearly identified together with the geometric variables:

1. Geometry constraints

- Length of the fuselage, set at 7.156 m
- Width, maximum 1.7m
- Interference with the intermeshing rotors

2. Aerodynamic objectives

- Positive lift coefficient, C_L
- Minimise drag coefficient, C_D
- Positive pitching moment coefficient, C_M

3. Geometric variables

- Upper and lower fuselage curvature
- Frontal shape: curved or flat
- Pylon covers height and shape

The objective of this analysis was to obtain the best shape which would comply with the geometric constraints provided in the initial sizing process and at the same time satisfy as closely as possible the aerodynamic parameters' aims. The internal systems layout is shown in figure 4d.

3.2.2 Three Configurations Considered

The chosen three configurations have been generated in collaboration with the CAD expert in the structures team [3]. Each one complied with the geometric constraints and was designed with careful consideration of the aerodynamic objectives while changing the geometric variables, as shown in figure 4.

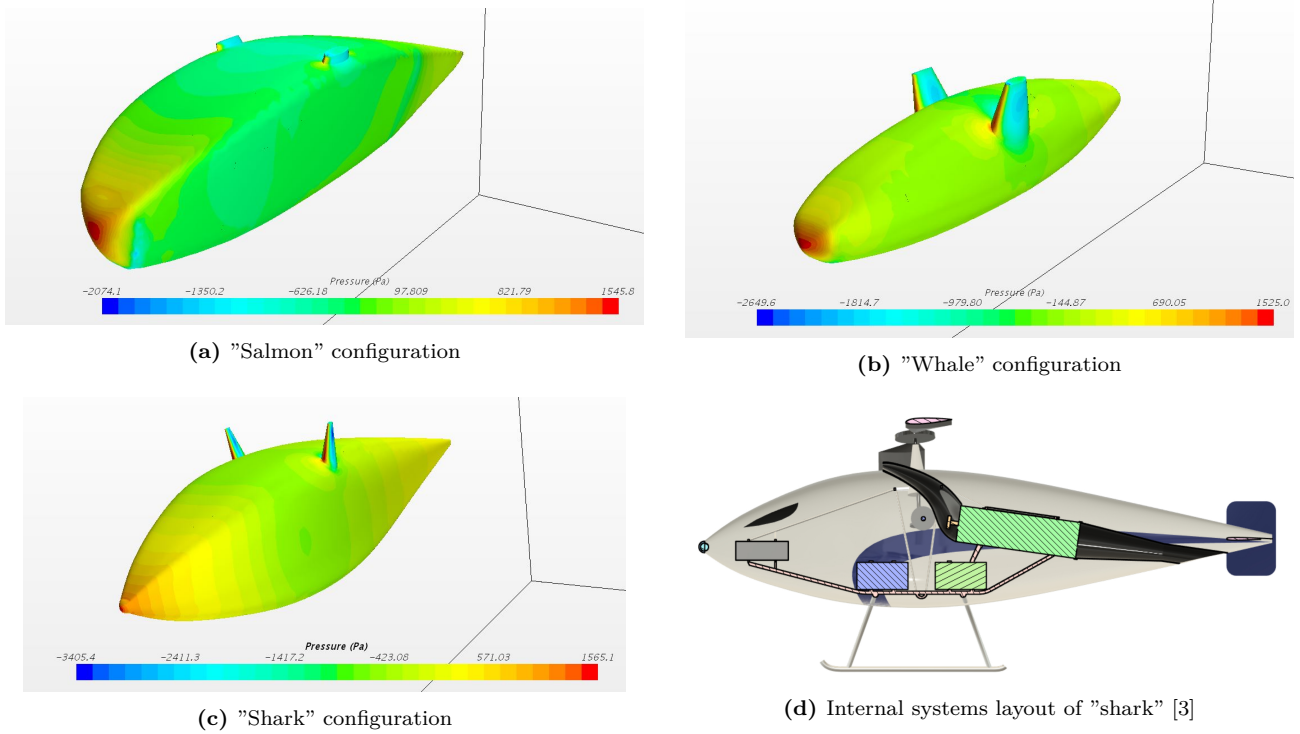


Figure 4: The three configurations analysed in CFD software StarCCM+

The "salmon" configuration was designed with a flat nose and a more squared shape rather than being rounded like a fuselage of an airliner. This was suggested to have a lower drag in forward flight [5]. The "whale" was created to investigate the effect of a droplet-like shape in forward flight. Lastly, the "shark" is a blend of the two bodies just described presenting a more streamlined shape but a squared rather than rounded front part. The other clear feature is the pylons' cover height: as described in [8], the rotor hub and shaft can contribute to up to 35 % of the total drag. Therefore, a **trade-off was found in increasing the volume of the fuselage while reducing the pylon's height**. This had a minimum value given by interference of the rotor blades, tilted at an angle of 14 degrees, with the fuselage.

3.2.3 CFD Modelling and Assumptions

Once the CADs were completed, the aerodynamic performance was computed with a widely used computational fluid dynamics software: StarCCM+. A trimmer volume mesh was implemented as this is most appropriate for large domains. The prism layer option also allowed to model the boundary layer forming on the fuselage surface. Another important feature was the volumetric mesh control, which allowed to refine the mesh near the fuselage and along the wake. A detail of such refinement for the "shark" is shown in figure 6. This refinement process has been consistently applied to all three shapes together with the models illustrated by the list over page.

- | | |
|----------------------|------------------------------------|
| 1. Three Dimensional | 5. Constant density |
| 2. Steady | 6. Turbulent |
| 3. Gas | 7. $k - \omega$ turbulence model |
| 4. Segregated flow | 8. Reynolds-averaged Navier-Stokes |

The segregated flow model was used due to the Mach number below 0.3 meaning that compressibility effects could be neglected (above 0.3 the coupled flow model would be more accurate). Regarding turbulence, the $k - \omega$ model was chosen over $k - \epsilon$ and Spalart-Allmaras. Indeed, the $k - \omega$ model is optimised to model turbulent flow close to the wall while $k - \epsilon$ is more appropriate for flows far from the solid boundary. Lastly, the SA model was deemed less relevant as it is primarily optimised for wings and aerofoil shapes [9], rather than larger and more bluff bodies such as fuselages.

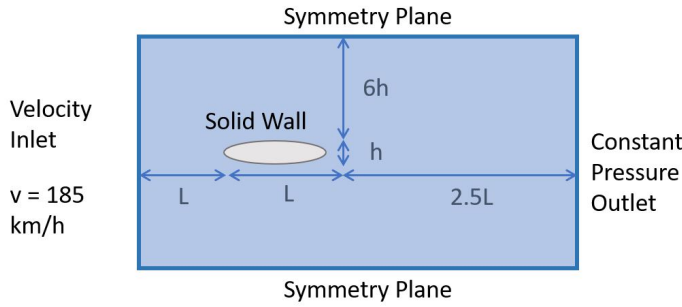


Figure 5: Boundary conditions and CV sizes (width = $2L$)

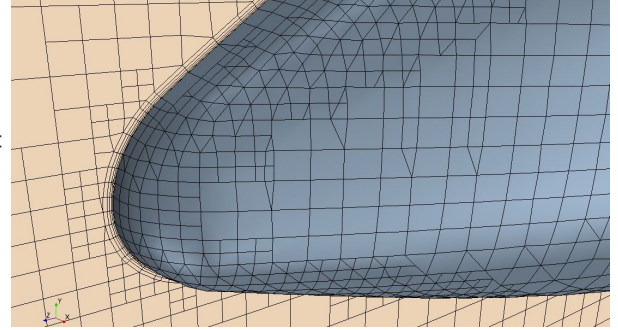


Figure 6: Mesh and prism layer nose view

As a last step, the boundary conditions aimed at simulating the flow properties in forward flight were selected. The left, right, top and bottom planes with symmetric conditions, the fuselage with non-slip solid wall, the velocity inlet at 185 km/h and constant pressure outlet. The primary control volume dimensions are illustrated in figure 5.

3.2.4 Results: Lift, Drag and Pitching Moment

The three fuselage shapes were tested and optimised at flow conditions corresponding to 1,500 ft. Such altitude was selected as the payload had to be captured, in the worst case scenario, at 3000 ft followed by a descent to 1500 ft and a cruise back to the boat at 185 km/h. The aerodynamic properties were considered at an angle of attack of 0, -6 and -12 degrees, as shown in figure 7. In particular, the pitching moment coefficient was computed from the centre of gravity location of the fuselage. Inspecting the results, the pitching moment vs alpha slope is positive ($\frac{dC_M}{d\alpha} > 0$), as expected from a streamlined fuselage shape. Furthermore, the drag coefficients of all three shapes gave figures of a similar order of magnitude, highlighting that the mesh and boundary conditions were applied consistently.

3.2.5 Final Design Choice

The final chosen design was the "Shark" configuration. Although the "salmon" shape performed best in drag and generated positive lift, it was discarded because of the issue of blade interference and camera placement. Increasing substantially the fuselage volume led to a noticeable reduction in pylon cover height, this would have had to be increased to avoid rotor interference, especially when the helicopter is not flying. As the blades are unloaded, they were likely to bend, making the initial starting rotation

when turning on the engines impossible. With further analysis, it was found that keeping the same shape while rising the pylon's height to avoid interference led to a larger drag and increased weight than the savings of drag and lift of the other two configurations. A discussion with the mechanics of flight and control team followed to understand the desired pitching moment [10]. The fact that a positive C_M was most ideal and that the "shark" fuselage consistently displayed a positive figure together with a lower drag and negative lift than the "whale", informed the final decision.

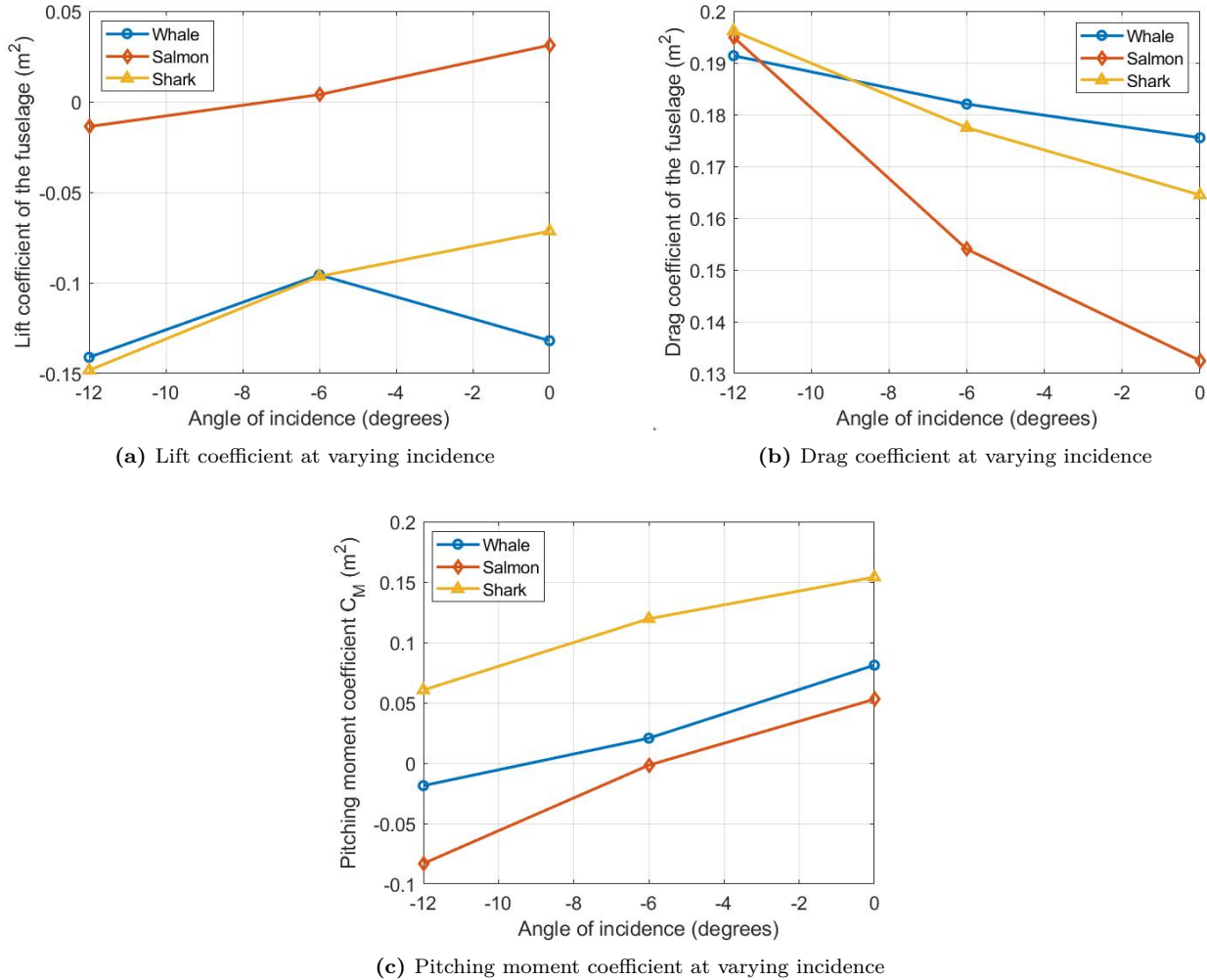


Figure 7: CFD results: lift, drag and pitching moment

The structures team was later arguing that the larger weight due to the higher volume of the fuselage was not worth the marginal reduction of drag of the "shark" over the "whale" [11]. However, after the weight of the material was investigated and the density was found to be around 0.8 kg/m^3 , the weight to drag saving ratio was: 0.33, meaning that the drag reduction was 3 times larger than the increase in weight. This highlighted that the "shark" was not only better from an aerodynamics and control perspective but it was also satisfying structural constraints and weight limits. This was the case because the skin of the fuselage was not carrying any load: it would only act as an aerodynamic cover. More generally, such design also required a collaborative effort with the propulsion team: while the main avionic systems and fuel tanks were unchanged, the turboshaft engine was inclined at 8 degrees from the horizontal axis to rise the maximum curvature requirement and further optimise the fuselage shape in forward flight.

3.2.6 Further Modifications and Improvements

Further research showed that the elliptically-shaped pylon covers produced a much larger drag than a faired geometry, so the modification was soon made, as suggested by Dr Montomoli. For the next iteration it would have been beneficial to camber slightly the bottom fuselage surface in order to reduce the effects of negative lift or even generating some, while keeping drag and pitching moments close to low and positive.

4 Root Aerofoil Choice

The design parameters and objectives were clearly identified at the beginning to ensure that the final design choice was informed by appropriately collected data for both the root and the tip aerofoil.

4.1 Design Parameters

The aerofoil at the root of the blade, being subjected to a low tangential speed and Mach number, allowed to neglect compressibility effects. In this regard, three design drivers were identified: stall speed, maximum L/D ratio and pitching moment. Five aerofoils were chosen with a large figure of C_{LMAX} to minimise stall while at the same time maintaining good performance with $(L/D)_{MAX}$. In addition, in order to minimise the rotor loads in forward flight, aerofoils with a close to zero or slightly positive pitching moment were screened for the final choice, as suggested in [12] and by Dr Ribera Vicent. As part of the five aerofoils, the SG6043 section was included. Nevertheless, this was soon neglected due to the extremely large and negative C_M of up to -0.18 which would induce too extreme loads on the blade.

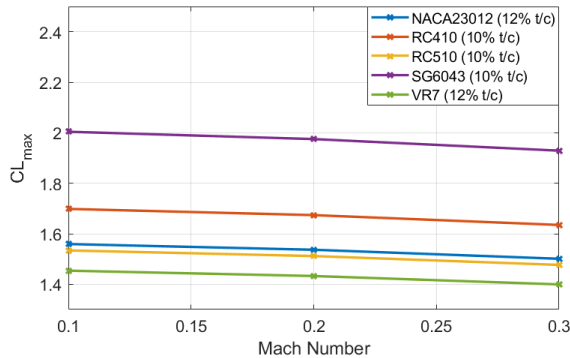


Figure 8: C_{LMAX} at different Mach numbers for 5 aerofoils

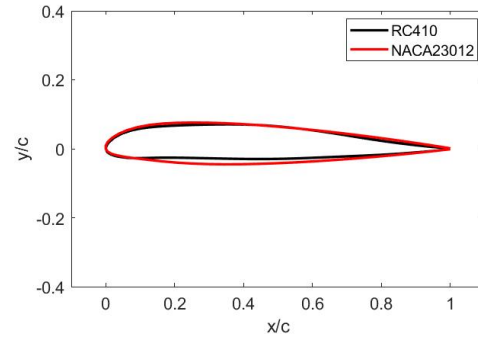


Figure 9: The final two aerofoils screened

Figure 8 displays the variation of C_{LMAX} at different Mach numbers: the best performing aerofoils, after excluding the SG 6043, were the RC 410 and NACA 23012. Considering C_M , both sections had figures very close to zero, reaching a maximum of 0.02 and a minimum of -0.04. The only distinctive feature was $(L/D)_{MAX}$, which was much larger for RC 410 having a value of 70 instead of 52 for NACA 23012.

4.2 Final Design Choice

The NACA 23012 aerofoil was eventually selected over the RC410. From an aerodynamics perspective, the RC410 would have been the optimal choice, being part of a family of aerofoils more optimised and refined than the NACA 4 series. However, after consulting the structures team it became clear that the maximum thickness to chord ratio of 10 % corresponding to RC 410 would lead to excessive loads on the blade [13]. Therefore, the final decision of selecting NACA 23012 was made with a t/c of 12 %, as shown in figure 9.

5 Drag Estimation

Drag was computed in detail for two conditions: hover and forward flight using both analytical methods and the results from the fuselage CFD testing.

5.1 Initial Drag Estimation

The helicopter equivalent flat plate area was first estimated using an analytical graph from Leishman [8] which relates airframe equivalent flat plate area f to gross weight for both utility and clean helicopters. The total equivalent flat plate area was found assuming a clean helicopter configuration for a gross weight of 2907 kg as seen in figure 10. Such initial estimation was not carried out for the payload capture mechanism assembly, figure 11, as the dimensions provided by the launcher team were not yet final and could vary significantly.

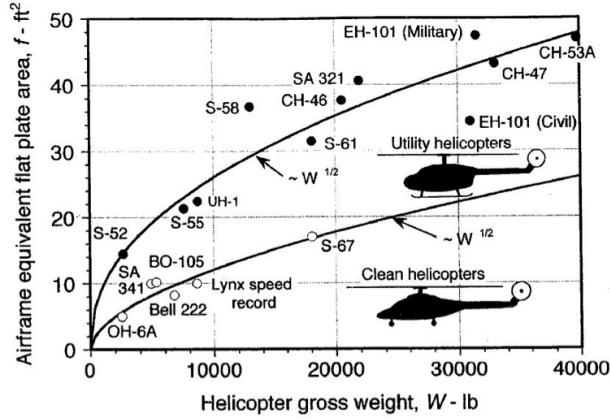


Figure 10: Equivalent flat plate area vs gross weight [1]

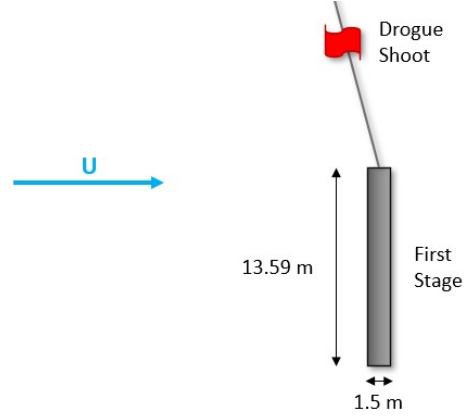


Figure 11: Slung load, drogue shoot configuration

The number obtained, $f = \frac{D}{\frac{1}{2}\rho u^2} = 1.57 \text{ m}^2$, was assumed to be an average of the airframe equivalent flat plate area of clean and utility helicopters. In fact, although the fuselage shape was designed to be clean from any equipment or weapons, the heavy-lifting requirement led to assume that the unmanned vehicle would be a combination of the two, hence the average. Although this assumption is a gross simplification, it was applied at the early design stage and especially during the weight iteration in collaboration with the structures and propulsion team [14]. Furthermore, this method served as an upper range since the unmanned vehicle was expected to have a much lower drag. In this regard, the equivalent total flat plate area f was usually 0.9 m^2 for clean helicopters and up to 4.5 m^2 for utility helicopters [1].

5.2 Detailed Drag Estimation

Once the components layout and general dimensions of the airframe were confirmed, it was then possible to estimate the detailed drag breakdown of each component of the helicopter. The calculations have followed closely the method described by Prouty [1], apart from some slight modifications. These were mainly regarding the fuselage drag prediction: the designed shape was not conventional meaning that a graph relating C_D to fineness ratio (fuselage length to maximum diameter) was not deemed appropriate. Thus, the CFD results from the fuselage optimisation were used instead, which led to a smaller value of 0.0314 instead of 0.04 at zero incidence. Table 2 illustrates the drag breakdown excluding the payload, which is discussed in the next section.

Component	Drag Coefficient	Area (m ²)	Flat Plate Area f (m ²)	Percentage (%)
Fuselage	0.031	2.698	0.085	20.3
Engine Inlet	0.090	0.125	0.011	2.69
Hub	0.450	0.172	0.114	27.2
Shaft	0.300	0.006	0.003	0.60
Skids	0.400	0.065	0.026	6.22
Rotor-Fuselage	0.022	2.698	0.059	14.2
Horizontal Stabiliser	0.024	0.209	0.005	1.19
Vertical Stabiliser	0.018	0.1369	0.002	0.44
Exhaust Drag	—	—	0.043	10.4
Miscellaneous Drag	—	—	0.0005	0.12
Total (with additional 20 %)	0.418			

Table 2: Detailed final drag breakdown, without payload (largest % contribution in bold)

The results displayed consistent ratios with past helicopter designs as the hub and fuselage contributed to the majority of drag. Interestingly, the exhaust drag was rather high compared to more conventional configurations. However, as the fuselage was small and the engine very large compared to its size due to high lifting requirements the figure was judged to be reasonable. In addition, the horizontal stabiliser was found to have a much larger flat plate area due to the significant impact of the rotor wake, always interacting with the tail as decided in the sizing procedure. Lastly, the skids were designed with a faired tubular section for the vertical structural members linking the fuselage, this reduced the drag coefficient by three times as verified by wind tunnel data [4].

5.3 Rotor-Fuselage Interaction and Hover Drag

The vertical drag in hover was also computed to fully analyse the performance of the helicopter. The induced velocity just underneath the rotors were found by my colleagues in the aerodynamic team and then processed by myself, as shown in figure 12 and 13. This was used to derive the velocity at a normalised vertical distance $z/R = 0.3$ with a development factor of 1.3. The dynamic pressure ratio could then be derived and lastly the vertical drag as a percentage of gross weight, following the procedure in Prouty [1].

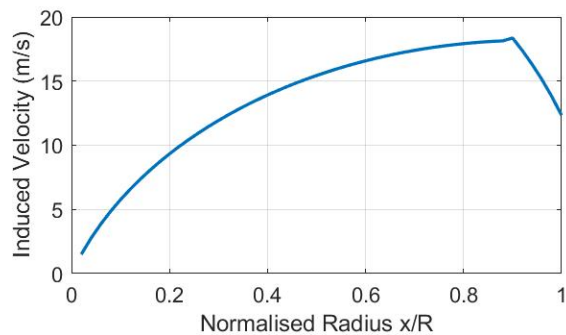


Figure 12: Induced velocity just underneath the rotors

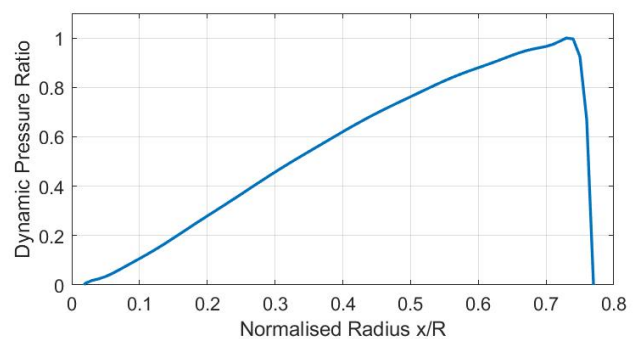


Figure 13: Dynamic pressure along the radius at $\frac{z}{R} = 0.3$

In specific, the total vertical drag contribution consisted of dividing the helicopter area seen from above into different sections having specific C_D coefficients of up to 0.9, known from wind tunnel testing [1]. During hover, the vertical drag to gross weight was $\frac{D_v}{GW} = 9.35\%$. The only significant assumption made consisted of keeping a constant distance between the fuselage and the rotor, meaning that an average was taken instead of integrating over the curved profile. In forward flight, the rotor-fuselage interaction was modelled and it has already been included in the detailed drag breakdown in table 2.

6 Slung Load Aerodynamic Modelling

The slung load modelling was carried out with the systems team to analyse the vehicle dynamics after the capture of the first stage rocket [15]. Such object was modelled as a cylinder with a diameter of 1.5 m and a length of 13.59 m, (refer to figure 11). The parafoil allowing the first stage to descend at a constant glide slope was studied after the capture manoeuvre as a smaller reefed body similar to a drogue shoot.

6.1 First Stage Modelling

The drag coefficient of a cylinder varies significantly with its aspect ratio (length divided by diameter L/D). It was found that the drag coefficient of an infinitely long cylinder for a Reynolds number of $Re = 8.8 \times 10^4$ was 1.2 and could go down up to around 0.6, as highlighted by Zdravkovich et al's wind tunnel testing [16]. More recently, Wenju et al have verified the wind tunnel data computationally and illustrated the variation of drag coefficient with Reynolds number at a fixed aspect ratio of 1.5 [17]. A similar procedure was followed to first verify the computational method where the results of StarCCM+ and wind tunnel data from [16] were compared. Secondly, the same StarCCM+ mesh constraints could be applied to a cylinder of $(L/D) = 9.06$ having the exact dimensions of the first stage. The CFD results together with wind tunnel data are shown in figure 14 with an average error of 2.6 %. The main simplification made consisted of assuming that the rotor's wake interaction with the cylinder could be neglected due to the rather long cable. This meant that the only component of velocity was horizontal.

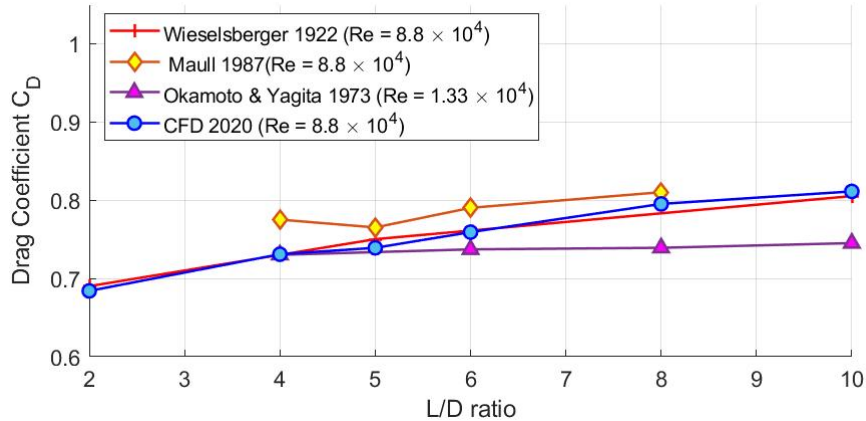


Figure 14: Wind tunnel data and CFD results

Once the first step was finalised, the first stage cylinder was introduced in the CFD environment keeping the same mesh structure: prism layers, and trimmer volume. As for the fuselage, volumetric control blocks allowed to refine the mesh in critical regions, namely the wake and the edges where viscous effects were most significant. Finally, the drag coefficient for the first stage at $Re = 2 \times 10^6$ corresponded to $C_D = 0.821$ while for the 2nd stage with $(L/D) = 5.1$, $C_D = 0.768$.

6.2 Drogue Shoot Modelling

The drogue shoot was not modelled with CFD as the exact shape was unknown and strictly dependent on the dynamics of the retracting mechanism. Therefore, an estimate was taken from a drogue shoot producer "Rocketman" with similar descent rate as the helicopter during capture. The manufacturer indicated a value of $C_D = 0.97$, which was used in the analysis to find the overall tension in the cable.

7 L/D Performance Estimation

The lift to drag ratio performance calculation consisted of collecting all the results from the drag, fuselage and blade analysis to verify the helicopter's behaviour at varying speed and altitude. As illustrated by Filippone [18], L/D can be expressed as $\frac{L}{D} = \frac{UW}{P}$ where P is the power required at varying speeds and altitudes, U the cruise speed and W the gross weight. The power required in cruise at any given speed without payload was derived by my colleagues in the aerodynamics team, while the additional power necessary with the payload was added after the slung load analysis. This could be done by computing the horizontal component of tension at any particular cruise speed, resulting in figure 15 and 16.

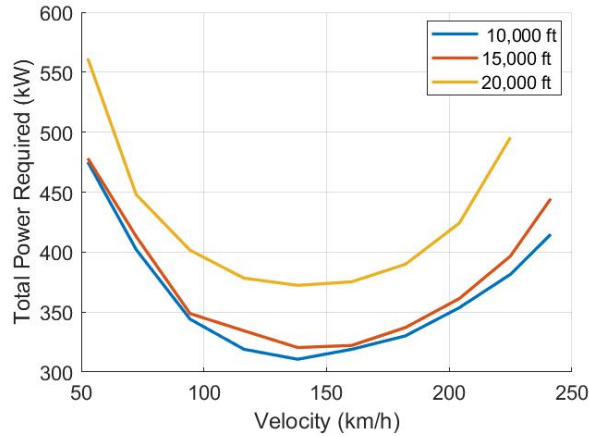


Figure 15: Required Power at varying speeds and altitudes

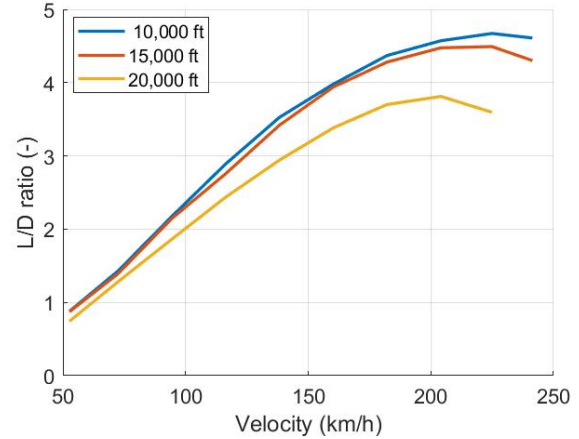


Figure 16: L/D ratio at varying speeds and altitude

8 Conclusion

The design project of an air vehicle capable of in air capture of a first stage rocket was completed by considering the most appropriate air vehicle, fuselage sizing optimisation, airfoil choice for the root of the blade and drag modelling with and without the payload. Although the models implemented displayed some limitations, namely the estimation of rotor-fuselage interaction with analytical rather than computational methods, the analysis gave reasonable consistent with previously designed helicopters. In particular, the $C_D = 0.0314$ of the chosen fuselage configuration was lower and not far from the analytical figure of 0.04. A more consistent optimisation could have been performed as described in [19], but due to the computational power and time constraints it was not possible. Lastly, this project has been particularly interesting as a specific vehicle configuration was not assigned at the beginning. The air vehicle choice has been the result of a week of arguments and debates for the best selling points vis-à-vis the information provided by the launcher team. Indeed, discussing constructively with other team members to reach a design compromise was one of the aspects that made the group design project unique and rewarding despite the current COVID-19 world crisis with virtual work settings.

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